

Mohammed Allabani

AI Engineer | Computer Vision & MLOps

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AI Engineer specialising in machine learning, deep learning, computer vision and multimodal AI. Experienced in taking models from dataset creation through training and optimisation to deployment on device, on the edge and in the cloud, using Python, PyTorch, TensorFlow, YOLO and Azure. B.Sc. in Artificial Intelligence Engineering (2026). Native Arabic speaker based in Jeddah.

PROJECTS

AI-Driven Object Tracking System

May – Jun 2026

End-to-end embedded vision system — ESP32-CAM, YOLO, Python

- Built a real-time object-tracking system end to end, spanning dataset creation, 3D part design, firmware development and model deployment.
- Curated and labelled a custom dataset of 2,000+ images for robust detection across varying lighting conditions and viewing angles.
- Diagnosed a 2 FPS inference ceiling on the ESP32-CAM and re-architected the pipeline — offloading YOLO inference to an external host and replacing a saturated 1 Mbaud serial link with Ethernet — reaching real-time framerates with high tracking precision.

AI-Based Adaptive Traffic Management System

Feb – May 2026

Six-person capstone — SUMO/CARLA, YOLOv11m, FastAPI, Next.js

- Owned the perception-analytics and synthetic-data workstream on a camera-driven adaptive traffic-signal system that cut intersection wait times by 40–54% against a fixed-time baseline.
- Developed a CARLA synthetic-data capture pipeline with YAML-configured spawn ratios, camera placements and weather, producing frame-accurate ground-truth labels and automated class-balance checks; this dataset trained a custom seven-class detector reaching 0.949 mAP50 on held-out synthetic data.
- Implemented a homography-based analytics layer (4-point ground-plane calibration) for lane-level speed estimation, queue counting, and red-light, chevron and highway-entry violation detection from monocular camera feeds.

EXPERIENCE

AI Intern — EARtech IT Company

Jun – Sep 2025

- Completed an intensive internship focused on AI, Azure Cloud and web development.
- Collaborated on Figma prototypes and built secure full-stack applications.
- Applied machine learning techniques and cloud deployment across real-world client projects.

VOLUNTEERING & LEADERSHIP

Student Community Representative — Bahçeşehir University

Oct 2025 – Jun 2026

- Represented 400+ students in official university discussions, academic committees and administrative meetings.
- Secured sponsorships from external partners for community events, going beyond typical student leadership responsibilities.
- Organised and led cultural events and community initiatives promoting intercultural dialogue, coordinating logistics and communication across diverse student groups.

SKILLS

Programming: Python, Swift, C++, R, SQL

ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Core ML

Computer Vision: YOLOv11, OpenCV, MediaPipe, ViT, Gemini Vision

Generative AI: AWS Bedrock, RAG, Prompt Engineering

Cloud & DevOps: Azure, Docker, Kubernetes, Supabase, CloudKit

EDUCATION

B.Sc. in Artificial Intelligence Engineering — Bahçeşehir University

2022 – Jun 2026

Istanbul, Turkey

CERTIFICATES

Advanced Concepts and Applications of Artificial Intelligence — SDAIA — Million Saudi Into AI Initiative

Explainable AI (XAI) Specialization — Duke University, Jan 2026

MLOps Platforms: Amazon SageMaker & Azure ML — Duke University

Deep Learning with PyTorch, Keras & TensorFlow — Professional Certificate — IBM, May 2025

AI Agents & MLOps for Production-Ready AI — Packt

Neural Networks & Deep Learning — DeepLearning.AI

AWS AI Practitioner Challenge — Udacity

LANGUAGES

Arabic — Native • English — Professional • Turkish — Intermediate